

AKHILESH B S

XR Simulation & Digital Twin Engineer

Unreal Engine 5 · C++ · Real-time Systems · XR

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PROFESSIONAL SUMMARY

XR Simulation & Digital Twin Engineer with 2.5+ years of experience developing real-time systems using Unreal Engine 5 (C++ & Blueprints). Specialized in flight simulation, architectural digital twins, XR applications, and cross-device control systems with live data integration through REST APIs, UDP, JSON, Google Sheets, and WebSocket-based communication pipelines.

TECHNICAL SKILLS

Core Technologies: Unreal Engine 5, C++, Blueprints, XR (AR/VR/MR), Digital Twin Systems

Simulation & Systems: JSBSim, UDP/TCP Networking, WebSockets, VRPN, REST APIs, JSON, Real-time Data Integration

Procedural & Environment: PCG Framework, Geo-Spatial Data, Dynamic Weather Systems

Visualization & UI: UMG, Pixel Streaming, Android APK UI Systems, Cross-device Control Interfaces, Lumen, Nanite, Sequencer, 4K Rendering

Platforms & Tools: Meta Quest, Varjo, Android, Blender, Git/GitHub, Python, Unity

WORK EXPERIENCE

Unreal Engine Developer | Credvest · Full-time | Oct 2025 – Present

- Architected Digital Twin and Architectural Visualization applications in Unreal Engine 5.
- Designed a real-time inventory system integrated with Google Sheets and CSV data.
- Built weather-driven digital twin environments using live weather APIs.
- Engineered WebSocket-based real-time control systems enabling Unreal Engine applications to be operated through tablets, Android APK interfaces, and external devices.
- Developed application analytics and tracking systems to monitor user interactions, button clicks, session activity, and feature usage, with structured data storage and processing in JSON format.
- Delivered optimized Android and VR builds with performance profiling and runtime optimization.
- Authored scalable Blueprint and C++ systems for production workflows.

Junior Unreal Engine Developer | LTX-LaunchTrax · Full-time | Sep 2024 – Oct 2025

- Engineered full-featured flight simulation systems integrating JSBSim physics over UDP, supporting aircraft dynamics, carrier-based operations, multiplayer missions, and air-to-air missile systems.
- Built large-scale procedural environments using Unreal Engine PCG with geo-spatial latitude/longitude data, dynamic terrain generation, weather systems, and scalable environment automation.
- Rigged and animated multiple aircraft models, developed cockpit UI/HUD systems, and implemented VRPN-based synchronization pipelines between Unreal Engine and Unity for distributed real-time simulation workflows.
- Delivered optimized VR simulations for Meta and Varjo headsets with real-time networking, dogfight simulations, and carrier landing scenarios.

Game Developer | LTX - LaunchTrax · Internship | Jul 2024 – Aug 2024

- Developed helicopter HUD systems and flight simulation workflows using JSBSim, rapidly contributing to production-level systems within weeks of joining.
- Converted JSBSim into a reusable Unreal Engine plugin, streamlining future simulation integration workflows and contributing to receiving a full-time offer during the internship period.

KEY PROJECTS

Flight Simulation: JSBSim Integration: Developed a real-time flight simulation system with aircraft rigging, cockpit HUDs, multiplayer workflows, and flight dynamics using JSBSim.

Architectural Digital Twin Application: Built a real-time architectural visualization platform with live inventory tracking, Google Sheets integration, and dynamic environment systems.

XR Hospital Interaction System: Designed immersive XR workflows for patient monitoring, interaction systems, and real-time data visualization.

Real-Time Communication & Control System: Implemented UDP and WebSocket-based communication pipelines with JSON data exchange, enabling real-time application control through Android APK and external devices.

Runtime Modeling Tool: Developed runtime 3D editing tools with undo/redo, transform controls, duplication, and save/load systems similar to Blender workflows.

EDUCATION

B.Tech in Mechanical Engineering | 2019 – 2022

SV College of Engineering and Technology · Score: 85%

Diploma in Mechanical Engineering | 2015 – 2019

Kuppam Engineering College · Score: 68%

CERTIFICATIONS & TRAINING

AR/VR Experiences - Ajna Creator | 6-Month Intensive Certification

• XR Interaction • Simulation Systems • Digital Twin Development • 3D Environment Creation

Full Stack Developer - JSpiders | 6-Month Training Certification

• Java • HTML • CSS • JavaScript • React • SQL